



**ASSESSMENT OF KNOWLEDGE AND ATTITUDE OF AGRICULTURAL
EDUCATION STUDENTS' ON CLIMATE CHANGE IN TAI SOLARIN
COLLEGE OF EDUCATION, OMU-IJEBU, OGUN STATE, NIGERIA**

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ABSTRACT

Climate change is one of the most fundamental challenges confronting humanity, its adverse impact is already evidence and may intensify exponentially overtime if nothing is done. This study examines the knowledge and attitude of agricultural education towards climate change. Data were collected from 48 agricultural education students purposively selected that cut across different levels. Constraint encountered in the course of the study were non-submission of the questionnaire by some of the respondents and high cost of transportation due to subsidy removal on petrol. Data were analyzed using both descriptive and inferential statistics such as frequency, percentages and Pearson Product Moment correlation coefficient (PPMC). The results revealed that (66.7%) of the respondents were 20-25 years of age, with majority (52.1%) were male. The result of Agricultural students' knowledge on climate change (21(43.8%) of the respondents did not consented that agricultural activities such as crop and livestock production contribute to climate change. Majority (47.3%) agreed that, if there was collective action the effects of climate change could be reduced. There was no significant relationship between socio-economic characteristics of respondents and their knowledge of climate change (Age of respondents ($r=.241$, $P>0.05$). Respondents sex ($r=-0.54$, $P>0.05$). Tribe ($r=.110$, $P>0.05$). Level of study ($r=.268$, $P>0.05$). Respondents Religion ($.221$, $P>0.05$) and Marital Status ($.071$, $P>0.05$). However, there was significant relationship between the level of study of the respondents and their attitude towards climate change ($r = .330^*$, $P < 0.05$). Based on the results of this study, it can be concluded that respondents' socio-economic characteristics had no significant relationship with their attitudes as well as their knowledge of climate change except their levels of study which had significant relationship with their attitudes toward climate change. It was therefore recommended that courses on climate change should be introduced at various levels of study in N.C.E curriculum.

INTRODUCTION

The intergovernmental Panel on Climate Change (IPCC, 2007; IPCC, 2014) defined climate change as statistically significant variation on climate that persists for an extended period, typically decades or longer. It includes shifts in the frequency and magnitude of sporadic weather events as well as the slow continuous rise in global mean surface temperature. Climate change is change in climate that is attributable directly or indirectly to human activities (Adebayo *et al.*, 2011). It affects the atmospheric conditions of the earth thereby leading to global warming. Climate change has the potential to affect all natural system thereby becoming a threat to human development and survival socially, politically and economically (Adebayo *et al.*, 2011).

At present, there is paucity of research on college of education students' climate change related threat perception, knowledge, or behaviours as vast majority of past studies focused on measuring these variables among primary and secondary school students (Falaye and Okwilagwe (2016). Among these studies, there is a consensus of reporting a general pattern of confusion about causes and consequences of climate change. Oluk and Oluk (2007) analysis on the topic of climate change discovered that over 50 percent of students at the university had the misconception that increase in temperature is majorly due to greenhouse effect or global warming, moreover, many of the common misconceptions observed in studies of primary and secondary school students on climate change also are found among University students (Freije *et al.*, 2017).

Considering the fact that the students today will be faced with task of making important climate change policy decisions and assume the responsibility of shaping the way our world addresses pressing climate change issues. Further studies are necessary in order to have an insight into how misconception about climate change concerns could be solved by students of higher institutions According to Ojomo *et al.*, (2015) Decision makers, invariably future decision makers should, thus, be knowledgeable about climate change to ensure implementation of appropriate policies and programmes that minimize the adverse effects and maximize the beneficial impacts of climate change.

Although knowledge does not always translate to action. Moreover, the study is germane at this point because the curriculum being implemented for NCE programme in Agricultural Education department in College of Education curriculum by National Commission for College for Education (NCCE, 2012)) presently has only one course on climate which is Agro-Climatology (AGE 115). The course outlines are as follows; Meaning and Scope of Agro-climatology; Equipment used in the study of Climatic Elements/factors; Climatic factors (Temperature, Precipitation, Wind, Solar radiation etc.) Temperature, Precipitation, Wind, Solar radiation, Ecological zones in Nigeria and their effects on ecological distribution of crops, livestock and Soil formation, Principles underlying weather forecasting. From the foregoing it means there is no specific topic on issues of climate change which further implies that students might go through the programme without any knowledge of climate change, climate change education programme is vital for equipping students with basic and advanced skills and knowledge and for influencing their behaviour and lifestyle, resulting in actions to adapt and mitigate climate change impacts throughout their lifetime that can also influence future generations (Choi *et al.*, 2017; United Nations Educational, 2010; United Nations, 1992). Climate change education can also create a path for students to initiate climate change activism and work with various actors (Cordero *et al.*, 2020). According to Filho *etal.*, (2023) despite the importance of climate change education in responding to the impacts of climate change, less attention has been given to systematically assessing the attitude, perceptions, and practices of students and the integration of the climate topic in the higher education institutions' curricula and co-curricular activities in a way that may guide changes in the curriculum and teaching practices, with the intent that new insights gained

from the research could be helpful to the design and redesign of curricular contents on climate change education. thus, the need to carry out this study on Knowledge and Attitude of Agricultural Students in Tai Solarin College of Education.

Objective of the study

Broad objective

The broad objective of the study was to assess the knowledge and attitude of agricultural education students on climate change

Specific Objectives

The specific objectives were to;

Describe the socio-economic characteristics of the students,

Assess the students' knowledge level of climate change

Assess the students' attitude towards climate change

Hypotheses

H₀₁; There was no significant relationship between the socio economic characteristics of the students and their attitude towards climate change

H_{a1}; There was significant relationship between socio economic characteristics of students and their attitude towards climate change

H₀₂; There was no significant relationship between students' socio economic characteristics and their knowledge of climate change

H_{a2}; There was significant relationship between students' socio economic characteristics and their knowledge of climate change

Methodology

Study Area

The study was conducted in Tai Solarin College of Education, Omu-Ijebu, located in Odogbolu Local Government area of Ogun –State-Nigeria. The campus at approximately between Latitude 742006.97N and 740410.109N and Longitude 593688.282E and 595636.462E. It is within the boundary of Ogun and Lagos States, and accessible through Ala-Omu road to Lagos State boundary (Wikipedia, 2019).

Research Design

The research design adopted was descriptive survey.

Sampling Procedure and Sample size

Multi-stage sampling technique was adopted to select sample for the study. A total of 48 Agricultural Education students were selected for this study. To achieve this selection, the following steps were taken;

The first stage involved stratification of students into their various levels as students in NCE1(14), students in NCE11(20), students in NCE111(14).

In the second stage, purposive sampling technique was used to select all the students from each level because they are relevant to the study for assessing students 'knowledge and attitude towards climate change.

As a result, fourteen (14) students NCE1, twenty (20) students NCE11, fourteen (14) students NCE111 in Agricultural Education department were chosen respectively. Therefore, the sample size of the study was forty-eight (48).

Table1: Sampling Frame and size of the Study

Tai Solarin College of Education	Session	Level	Total Number of Students	Sample Size
	2021/2022	NCEI	14	14
		NCE11	22	20
		NCE111	14	14
Total			50	48

Source; Preliminary field survey, 2022

Data Collection

A well-structured interview guide was designed and used to elicit information from the respondents. Knowledge of students of climate change was dichotomous variables measured at nominal scale as true/false while the attitude of students towards climate change was measured on 4-point rate scale as; SD: Strongly disagree, D: Disagree, A: Agree, SA; Strongly Agree.

Data Analysis

Data obtained for the study were subjected to descriptive and inferential statistics using Statistical Package for Social Science (SPSS) version 15.0. Descriptive statistics such as frequency counts, percentage and mean were used to analyze and achieve the objectives i, ii, iii of the study. Pearson product moment correlation coefficient (PPMC) was used to analyze the hypotheses of the study.

Pearson' Product-Moment Correlation Coefficient

It is about the most widely used measures of association for interval and ratio scales. It measures linear association between interval variables. The data should be interval (e.g. age) or ratio (years spent in school). However, ordinal variables can be correlated.

$$\bar{x} = \frac{1}{n} \sum_{c=1}^N x_i$$

$$\bar{y} = \frac{1}{N} \sum_{c=1}^N y_1$$

$$R_p = \sum_{c=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\sqrt{\sum_{c=1}^n (x_1 - \bar{x})^2 \sum_{c=1}^n (y_1 - \bar{y})^2}$$

where $\bar{x}$ and $\bar{y}$ are sample mean of $x_1, x_2, \dots$ and $y_1, y_2, \dots, y_n$ respectively.

Results and Discussion

Table 2: Distribution of respondents according to their Socio-economic Characteristics

S/NO	Variables	Frequency	Percentage
1	Age (years)		
	15-20	16	33.3
	20-25	32	66.7
	Total	48	100
2	Sex		
	Male	25	52.1
	Female	23	47.9
	Total	48	100
3	Tribe		
	Hausa	01	2.1
	Igbo	04	8.3
	Yoruba	43	89.6
	Total	48	100
4	Level		
	NCE1	14	29.2
	NCE2	20	41.6
	NCE3	14	29.2
	Total	48	100
5	Religion		
	Muslim	17	35.4
	Christian	30	62.5
	Traditional	01	2.1
	Total	48	100
6	Marital status		
	Married	02	4.2
	Single	46	95.8
	Total	48	100

Source: Field Survey, 2022

Socio-economic Characteristics of respondents in Table2 revealed that 16 (33.3%) of the respondents were 15-20 years old, while 32(66.7%) of the respondents were age 20-25 years. This shows that the respondents fit in into the age bracket of students that normally run National Certificate of Education (NCE) programme in Colleges of Education in Nigeria.

Also, from the table there were more male respondents 25(52.1%) than their female counterparts 23 (47.9%) this is contrary to the notion that there are more female students in the Colleges of Education than male students or women are more in the teaching profession than their men counterparts. Another inference from this finding is that there are more male students studying Agricultural Education at Tai Solarin College Education, Omu-Ijebu. The reason for this finding is that farming activities or production tend to be dominated by the males while the females are more involved in marketing than production.

In terms of tribe, the Yoruba were in the majority 43(89.6%). This is as a result of the location of the College of Education which is Yoruba land.

The frequency distribution of respondents in terms of level of study indicated that NCE1 students were 14(29.2%), NCE11 students 20(41.6 %) and NCE3 14(29.2%) respectively. The possibly explanation for this development is the fluctuating and dwindling enrolment of students into NCE programme at the tertiary level of education in Nigeria. The result revealed that majority of the respondents 30(62.5%) were Christians. The table1 also revealed that majority respondents 46 (95.8%) were single

S/N	Items on Knowledge of Climate Change	Response	Frequency	Percentage
1.	Climate change is nothing but a natural fluctuation in temperature of the earth	True	43	89.6
		False	05	10.4
2.	Manifestation of climate change vary from one region to another	True	23	47.9
		False	25	52.1
3.	It is very late for the human community to do anything to stop climate change	True	31	64.6
		False	17	35.4
4.	Agricultural activities like crop and livestock production contribute to climate change	True	27	56.3
		False	21	43.8
5.	`Overall, climate change is bad, it is more harmful than beneficial	True	31	64.6
		False	17	35.4
6.	Human activities do not tangibly affect temperature of the surface of the earth	True	27	56.3
		False	21	43.8

7.	Climate change primarily resulted from ozone hole	True	26	54.3
		False	22	43.8
8.	The sector with the highest contribution to climate change is the transport sector	True	27	56.3
		False	21	43.8
9.	Environmental pollution resulting from the industry is the principal cause of climate change	True	24	50.0
		False	24	50.0
10.	Climate change contributes to increased water and wind erosion of the soil	True	27	56.3
		False	20	41.7
11.	Decline in yield livestock and crop production, thus, deterioration of food security may result from climate change	True	22	45.8
		False	26	54.2
12.	Climate change can result in shortage of water that is suitable for household use, irrigation for crop, livestock use	True	30	62.5
		False	18	37.5
13.	Climate change reduces incidence of infections of animal, plant and human diseases	True	26	54.2
		False	22	45.8
14.	Climate change reduces incidence of contagious animal, plant and human diseases.	True	22	45.8
		False	26	54.2

Table3: Distribution of Respondents according to their Knowledge on Climate Change

Source: Field Survey, 2022

Entries in Table 3 show the distribution of respondents according to their Knowledge on Climate Change. The result of Agricultural students' knowledge on climate change as indicated in table 3 revealed that most of the students 43(89.6%) accepted that climate change is nothing but natural fluctuation in temperature of the earth. This is similar to the result of Wang *etal.*, (2019) that the respondents were challenged in understanding data that forms the evidence for global climate change, whether this phenomenon is anthropogenic or natural, or both. Nearly 25(52.1%) of the respondents did not savvy that the manifestations of climate change vary from one climatic region to another, this is contrary to the findings of Filho *etal.*, (2023) that University students agreed that climate change varies from one region to the other, while some of them 17(35.4%) opined that it is very late for human community to do anything to mitigate or control climate change. More than half 27(56.3%) of the respondents consented that agricultural activities such as crop and livestock production contribute to climate change. Majority of the respondents agreed that human activities tangibly affect temperature of the surface of the earth 27(56.3%) These support findings of Filho *etal.*, (2023) that 70% of the university students attributed the cause of climate change mainly to human activity.

whereas 26(54.2%) of the respondents indicated that climate change primarily resulted from the ozone hole. This supports the assertion of Ezeudu (2016) that the concentration of the greenhouse gases in the atmosphere caused ozone layer depletion. At the same time, majority 27(56.3%) of the respondents agreed that the transportation sector is the highest contributor to climate change, it was found that the respondents could not decide whether or not environmental pollution resulting from industry is the principal cause of climate change while majority of the respondents 27(56.3) agreed that climate change contributes to increase water and wind erosion of the soil. Whereas 30(62.5%) of the respondents concurred that climate change could result in shortage of water that is suitable for household use, irrigation for crop production and livestock use. The possible explanation for the findings, is that at any rate climate change can result to excessive rainfall that can cause flooding, high rate of run off leading to erosion, it can also be accompanied by high wind velocity that can have destructive effects on human and farming activities. Two, it can result to sudden cessation of rainfall with accompanied drought or shortage of water or suitable water for household use, irrigation (calabash, bucket and watering can irrigation or smallholder farmer irrigation methods) for crop and livestock use. As a final point, the respondents' response 22(45.8%) was that climate change increases incidence of contagious animal, plant and human diseases. This is in line with the view of Ekezie (2010) that the rise in temperature as a result of climate change could lead to drought, desertification, declining water table, loss of some plant and animal species, low crop yield and outbreak of climate related diseases like malaria and meningitis. The inference from the mixed responses to items on the knowledge of the respondents on climate change is that the students were not taught any course on climate change. This confirms the issue raised in the problem statement that there is no specific topic on issues of climate change which further implies that students might go through the Agricultural Education programme without adequate or any knowledge of climate change. The result is also in line with findings of Filho *et al.*, (2023) that the student's knowledge of climate change risks varies across gender, age, and academic education. This also corroborates Akrofi *et al.*, (2019) 'students' involvement in climate change-related workshops and campaigns significantly influenced their knowledge levels and students' attitudes and behaviours are dependent on their level of awareness and knowledge about climate change issues. This in turn implies that their cognitive repertoire depends on their level of participation in climate change courses, workshops and campaigns, membership in environmentalist groups, and access to climate information

Table4: Distribution of respondents by their attitudes towards Climate Change

S/N	Items	Response	Frequency	Percentage
		SD	07	14.6
1.	Global warming does not give room for concern on climate change	D	15	31.3
		A	18	37.5
		SA	08	16.6
		SD	18	37.5
2.	The lack of concern for problem of climate change by some individuals, government bothers me	D	11	22.9
		A	05`	10.4
		SA	14	29.2
		SD	13	27.1
3	Scientists will discover solution to climate change, so we need not overly concern ourselves about it	D	07	14.6
		A	10	20.8
		SA	18	37.5
		SD	14	29.2
4.	Nature has a capacity to heal itself, so I do not worry about climate change	D	13	27.1
		A	07	14.6
		SA	14	29.2
		SD	07	14.6
5.	Climate change/global warming is a serious problem we are facing now	D	15	31.2
		A	18	37.5
		SA	08	16.7
		SD	03	6.3
6.	If collective action is taken, climate change effects will be reduced	D	08	16.7
		A	14	29.2
		SA	23	47.9
7.	Environmental problems associated with global warming can be solved	SD	05	10.4
		D	13	27.1

		A	17	35.4
		SA	13	27.1
		SD	13	27.1
8.	Whatever we do, we cannot slow down the warming of the earth due to greenhouse gases	D	13	27.1
		A	10	20.8
		SA	12	25.0
		SD	27	56.3
9.	Man 's activities do not change the climate	D	05	10.4
		A	09	18.8
		SA	07	14.6
		SD	27	56.3
10.	No amount of effort can be put in to change the phenomenon of climate change	D	05	10.4
		A	09	18.8
		SA	07	14.6
		SD	14	29.2
11.	Simple things can be done to slow down global warming or lessen its impact.	D	11	22.9
		A	14	29.2
		SA	09	18.8
		SD	16	33.3
12.	I plan to take some actions to stop global warming	D	06	12.5
		A	12	25.0
		SA	14	29.2
		SD	10	20.8
13.	I do not know enough about global warming, so more campaign must be undertaken before action can be taken	D	15	31.3
		A	07	14.6
		SA	16	33.3
14.	I would like to work with others to slow down global warming	SD	12	25.0

		D	06	12.5
		A	20	41.7
		SA	10	20.8
		SD	10	20.8
15.	I am willing to sacrifice my convenience to help reduce greenhouse gas emission	D	13	27.1
		A	12	25.0
		SA	13	27.1
		SD	14	29.2
16.	My activities have no significant contribution to global warming, so I do not intend to take actions to change them	D	15	31.3
		A	12	25.0
		SA	07	14.6

SD: Strongly Disagree, D: Disagree, A: Agree, SA; Strongly Agree

Field Survey, 2022.

Entries Table 4 show the distribution of respondents by their attitudes towards Climate Change. The result shown that majority of the respondents 18(37.5%) indicated that global warming did give room for concern on climate change, This implies the issue of global warming should give everybody cause for concern because there is gradually reduction in ozone layer which bring about various effects of climate change being experienced in world today, while 18(37.5%) on the contrary indicated that the lack of response by individuals and government towards the problem of climate change did not bother them. Majority 18 (37.5%) opined that scientist will discover solutions to the problem of climate change, most of them 18(37.5%) concurred that climate change/ global warming is a serious problem we are facing now, 23(47.9%) of them strongly agree that if there is collective action, we will be able to reduce the effects of climate change. About 17(35.4%) of the respondents agreed that they will be able to solve environmental problems associated with global warming. Most of the respondents 13(27.1%) strongly disagree that whatever they do, they cannot slow down the warming of the earth due to greenhouse gases. It means if appropriate actions are taken by stakeholders, greenhouse gas emission may be controlled. Majority of the respondents 27(56.3%) strongly disagree that man's activities do not change

the climate as well as no amount of effort can be put in to change it. It means human activities contribute to the problem of climate change and if appropriate climate smart programmes are put in place the phenomenon could be reversed particularly in relation to Agriculture or farming activities. Nearly 16(33.3%) plan to take some actions to stop global warming while 20(41.7%) would like to work with others to help slow down global warming. These imply that climate change mitigation and adaptation strategies should be a collective action to be undertaken by all parties involved in climate change actions. Some of the respondents 16(33.3%) concurred, not knowing enough about global warming, so more campaign must be undertaken before taken action. This emphasizes the need for the inclusion of climate change courses in the curriculum at the NCE programme so the effects can trickle down to the Basic level of education (i.e. Primary and Junior secondary schools). Also, sensitization on causes, effects and mitigation and adaptation strategies on climate change at the farm level. All of the above responses of the respondents are in line with the views of Shaman and Knowlton, (2017) that students' involvement and participation in climate change adaptation and mitigation activities are crucial to address the direct and indirect impacts of climate change. However, since students have not reached the desired level of consciousness, educational curriculums on climate change need to be designed to positively affect students' perceptions about the environment, and their courses should improve university students 'or tertiary institution students' scientific skills and knowledge that shape their attitudes and beliefs about climate change. This finding supports Oruonye (2011) who found out that tertiary institution students possessed low attitude towards climate change. Ishaya and Obaje (2008) also found out that secondary school students in Jema, Kaduna State, possessed low attitude towards climate. Majority of the respondents disagree 15(31. 3%) with the notion that their activities have no significant contribution to global warming so I do not intend to take actions to change them. It means that their activities contribute significantly to the problem of global warming and they are willing to take action to change the situation. The inference that can be drawn from the responses of the respondents above is that if the students were taught, given adequate orientation about the phenomenon of climate change at the NCE level or whatever level of education they will be able to tackle the scenario of climate change in the future and they are also willing to take positive actions to solve the problem of climate change. This is line with the view of Bozoglu *et al.*, (2022) that because these students will either be part of experts or general public in near future. This is corroborated by Molthan-Hill *et al.* (2019), that the role of the University (tertiary institutions) is important in climate change education in order for the future leaders to make informed decisions and for the public needs to be embedded in the climate change mitigation and adaptation tools and practices.

Table5: Correlation between socio - economic characteristics of respondents and their knowledge of climate change

Socio- economic characteristics	Correlation	Students' knowledge
	Pearson Correlation	.241
Age of respondents	Sig. (2-tailed)	.099
	N	48
	Pearson Correlation	-.054
Respondent Gender	Sig. (2-tailed)	.715
	N	48
	Pearson Correlation	.110
Respondents Tribe	Sig. (2-tailed)	.458
	N	48
	Pearson Correlation	.268
Level of study	Sig. (2-tailed)	.066
	N	48
	Pearson Correlation	.221
Respondents Religion	Sig. (2-tailed)	.131
	N	48
	Pearson Correlation	.071
Marital Status	Sig. (2-tailed)	.631
	N	48
	Pearson Correlation	1
Students' knowledge	Sig. (2-tailed)	.240
	N	48

$P > 0.05$ = Not significant but if $P < 0.05$ = shows that there is significant relationship

Source: Field Survey, 2022

The results in Table 5 show the correlation coefficients of the relationship between selected socio-economic characteristics and agricultural students' knowledge of climate change. There was no relationship between the selected socio-economic characteristics and students' knowledge of climate change such as age of respondents (.241, $P>0.05$). Respondents gender (- 0.54, $P>0.05$). Tribe (.110, $P>0.05$). Level of study (.268, $P>0.05$). Respondents Religion (.221, $P>0.05$) and Marital Status (.071, $P>0.05$). This implies that no topic on climate change was included in the Agricultural education curriculum as the students did not have any knowledge of climate change. This confirms the issue raised in the problem statement that there is no specific topic on issues of climate change which further implies that students might go through the Agricultural Education programme without adequate or any knowledge of climate change. The result is also in line with findings of Filho *et al.*, (2023) that the student's knowledge of climate change risks varies across gender, age, and academic education. This also corroborates Akrofi *et al.*, (2019) 'students' involvement in climate change-related workshops and campaigns significantly influenced their knowledge levels and students' attitudes and behaviours are dependent on their level of awareness and knowledge about climate change issues. This in turn implies that their cognitive repertoire depends on their level of participation in climate change courses, workshops and campaigns, membership in environmentalist groups, and access to climate information.

Table 6: Correlation between socio - economic characteristics of respondents and their Attitude towards climate change

Socio - economic characteristics	Correlation	Students Attitude
	Pearson Correlation	.178
Age of respondent	Sig. (2-tailed)	.226
	N	48
	Pearson Correlation	_.173
Respondents Gender	Sig. (2-tailed)	.239
	N	48
	Pearson Correlation	_.041
Respondents Tribe	Sig. (2-tailed)	.785

	N	48
	Pearson Correlation	.330*
Level of Study	Sig. (2-tailed)	.022
	N	48
	Pearson Correlation	_.082
Respondents Religion	Sig. (2-tailed)	.581
	N	48
	Pearson Correlation	_.026
Marital Status	Sig. (2-tailed)	.859
	N	48
	Pearson Correlation	1
Students Attitude	Sig. (2-tailed)	
	N	1104

Source: Field Survey, 2022 $P > 0.05$ = Not significant but if $P < 0.05$ = Significant

Table 6 also revealed that most of the respondents' socio-economic characteristics did not have any significant relationship with their Attitude towards climate change except their levels of study which show a significant relationship ($r = .330^*$, $P < 0.05$) with their Attitude. The inference that can be drawn from the responses of the respondents above is that if the students were taught, given adequate orientation about the phenomenon of climate change at the NCE level or whatever level of education they will be able to tackle or take proactive actions against the scenario of climate change in the future. The findings of Filho *et al.*, (2023) pointed out that universities (tertiary institutions) could improve student awareness by organising climate change-related themes in discussion sessions and incorporating the outcome into the curriculum and co-curricular activities.

Conclusion

Based on the results of this study, it can be concluded that respondents' socio-economic characteristics had no significant relationship with their attitudes as well as their knowledge of climate change except their levels of study which had significant relationship with their attitudes toward climate change.

Recommendations

Based on the results of this study, the following recommendations are hereby made:

1. Curriculum at the National Certificate of Education should be reviewed by making climate change part of the topic in Agro-climatology course which must be offered by all the students irrespective of their course of study in order to enlighten them on the effect of climate change.
2. Climate change awareness club should be established in higher institutions so as to improve students' knowledge and attitudes toward climate change, such clubs can be challenged to identify and find solution to the insurmountable problems of climate change.

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The intergovernmental Panel on Climate Change (IPCC, 2007; IPCC, 2014) defined climate change as statistically significant variation on climate that persists for an extended period, typically decades or longer. It includes shifts in the frequency and magnitude of sporadic weather events as well as the slow continuous rise in global mean surface temperature. Climate change is change in climate that is attributable directly or indirectly to human activities (Adebayo *et al.*, 2011). It affects the atmospheric conditions of the earth thereby leading to global warming. Climate change has the potential to affect all natural system thereby becoming a threat to human development and survival socially, politically and economically (Adebayo *et al.*, 2011).

At present, there is paucity of research on college of education students' climate change related threat perception, knowledge, or behaviours as vast majority of past studies

focused on measuring these variables among primary and secondary school students (Falaye and Okwilagwe (2016). Among these studies, there is a consensus of reporting a general pattern of confusion about causes and consequences of climate change. Oluk and Oluk (2007) analysis on the topic of climate change discovered that over 50 percent of students at the university had the misconception that increase in temperature is majorly due to greenhouse effect or global warming, moreover, many of the common misconceptions observed in studies of primary and secondary school students on climate change also are found among University students (Freije *et al.*, 2017).

Considering the fact that the students today will be faced with task of making important climate change policy decisions and assume the responsibility of shaping the way our world addresses pressing climate change issues. Further studies are necessary in order to have an insight into how misconception about climate change concerns could be solved by students of higher institutions According to Ojomo *et al.*, (2015) Decision makers, invariably future decision makers should, thus, be knowledgeable about climate change to ensure implementation of appropriate policies and programmes that minimize the adverse effects and maximize the beneficial impacts of climate change.

Although knowledge does not always translate to action. Moreover, the study is germane at this point because the curriculum being implemented for NCE programme in Agricultural Education department in College of Education curriculum by National Commission for College for Education (NCCE, 2012)) presently has only one course on climate which is Agro-Climatology (AGE 115). The course outlines are as follows; Meaning and Scope of Agro-climatology; Equipment used in the study of Climatic Elements/factors; Climatic factors (Temperature, Precipitation, Wind, Solar radiation etc.) Temperature, Precipitation, Wind, Solar radiation, Ecological zones in Nigeria and their effects on ecological distribution of crops, livestock and Soil formation, Principles underlying weather forecasting. From the foregoing it means there is no specific topic on issues of climate change which further implies that students might go through the programme without any knowledge of climate change, climate change education programme is vital for equipping students with basic and advanced skills and knowledge and for influencing their behaviour and lifestyle, resulting in actions to adapt and mitigate climate change impacts throughout their lifetime that can also influence future generations (Choi *et al.*, 2017; United Nations Educational, 2010; United Nations, 1992). Climate change education can also create a path for students to initiate climate change activism and work with various actors (Cordero *et al.*, 2020). According to Filho *etal.*, (2023) despite the importance of climate change education in responding to the impacts of climate change, less attention has been given to systematically assessing the attitude, perceptions, and practices of students and the integration of the climate topic in the higher education institutions' curricula and co-curricular activities in a way that may guide changes in the curriculum and teaching practices, with the intent that new insights gained from the research could be helpful to the design and redesign of curricular contents on

climate change education. thus, the need to carry out this study on Knowledge and Attitude of Agricultural Students in Tai Solarin College of Education.

Objective of the study

Broad objective

The broad objective of the study was to assess the knowledge and attitude of agricultural education students on climate change

Specific Objectives

The specific objectives were to;

Describe the socio-economic characteristics of the students,

Assess the students' knowledge level of climate change

Assess the students' attitude towards climate change

Hypotheses

H₀₁; There was no significant relationship between the socio economic characteristics of the students and their attitude towards climate change

H_{a1}; There was significant relationship between socio economic characteristics of students and their attitude towards climate change

H₀₂; There was no significant relationship between students' socio economic characteristics and their knowledge of climate change

H_{a2}; There was significant relationship between students' socio economic characteristics and their knowledge of climate change

Methodology

Study Area

The study was conducted in Tai Solarin College of Education, Omu-Ijebu, located in Odogbolu Local Government area of Ogun –State Nigeria. The campus at approximately between Latitude 742006.97N and 740410.109N and Longitude 593688.282E and 595636.462E. It is within the boundary of Ogun and Lagos States, and accessible through Ala-Omu road to Lagos State boundary (Wikipedia, 2019).

Research Design

The research design adopted was descriptive survey.

Sampling Procedure and Sample size

Multi-stage sampling technique was adopted to select sample for the study. A total of 48 Agricultural Education students were selected for this study. To achieve this selection, the following steps were taken;

The first stage involved stratification of students into their various levels as students in NCE1(14), students in NCE11(20), students in NCE111(14).

In the second stage, purposive sampling technique was used to select all the students from each levels because they are relevant to the study for assessing students 'knowledge and attitude towards climate change.

As a result, fourteen (14) students NCE1, twenty (20) students NCE11, fourteen (14) students NCE111 in Agricultural Education department were chosen respectively. Therefore, the sample size of the study was forty-eight (48).

Table1: Sampling Frame and size of the Study

Tai Solarin College of Education	Session	Level	Total Number of Students	Sample Size
	2021/2022	NCEI	14	14
		NCE11	22	20
		NCE111	14	14
Total			50	48

Source; Preliminary field survey, 2022

Data Collection

A well-structured interview guide was designed and used to elicit information from the respondents. Knowledge of students of climate change was dichotomous variables measured at nominal scale as true/false while the attitude of students towards climate change was measured on 4-point rate scale as; SD: Strongly disagree, D: Disagree, A: Agree, SA; Strongly Agree.

Data Analysis

Data obtained for the study were subjected to descriptive and inferential statistics using Statistical Package for Social Science (SPSS) version 15.0. Descriptive statistics such as frequency counts, percentage and mean were used to analyze and achieve the objectives i, ii, iii of the study. Pearson product moment correlation coefficient (PPMC) was used to analyze the hypotheses of the study.

Pearson' Product-Moment Correlation Coefficient

It is about the most widely used measures of association for interval and ratio scales. It measures linear association between interval variables. The data should be interval (e.g. age) or ratio (years spent in school). However, ordinal variables can be correlated.

$$\bar{x} = \frac{1}{n} \sum_{c=1}^N x_i$$

$$\bar{y} = \frac{1}{N} \sum_{c=1}^N y_1$$

$$R_p = \sum_{c=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\sqrt{\sum_{c=1}^n (x_1 - \bar{x})^2 \sum_{c=1}^n (y_1 - \bar{y})^2}$$

where $\bar{x}$ and $\bar{y}$ are sample mean of $x_1, x_2, \dots$ and $y_1, y_2, \dots, y_n$ respectively.

Results and Discussion

Table 2: Distribution of respondents according to their Socio-economic Characteristics

S/NO	Variables	Frequency	Percentage
1	Age (years)		
	15-20	16	33.3
	20-25	32	66.7

	Total	48	100
2	Sex		
	Male	25	52.1
	Female	23	47.9
	Total	48	100
3	Tribe		
	Hausa	01	2.1
	Igbo	04	8.3
	Yoruba	43	89.6
	Total	48	100
4	Level		
	NCE1	14	29.2
	NCE2	20	41.6
	NCE3	14	29.2
	Total	48	100
5	Religion		
	Muslim	17	35.4
	Christian	30	62.5
	Traditional	01	2.1
	Total	48	100
6	Marital status		
	Married	02	4.2
	Single	46	95.8
		48	100

Source: Field Survey, 2022

Socio-economic Characteristics of respondents in Table2 revealed that 16 (33.3%) of the respondents were 15-20 years old, while 32(66.7%) of the respondents were age 20-25 years. This shows that the respondents fit in into the age bracket of students that normally run National Certificate of Education (NCE) programme in Colleges of Education in Nigeria.

Also from the table there were more male respondents 25(52.1%) than their female counterparts 23 (47.9%) this is contrary to the notion that there are more female students in the Colleges of Education than male students or women are more in the teaching profession than their men counterparts. Another inference from this finding is that there are more male students studying Agricultural Education at Tai Solarin College Education, Omu-Ijebu. The reason for this finding is that farming activities or production tend to be dominated by the males while the females are more involved in marketing than production.

In terms of tribe, the Yoruba were in the majority 43(89.6%). This is as a result of the location of the College of Education which is Yoruba land.

The frequency distribution of respondents in terms of level of study indicated that NCE1 students were 14(29.2%), NCE11 students 20(41.6 %) and NCE3 14(29.2%) respectively. The possibly explanation for this development is the fluctuating and dwindling enrolment of students into NCE programme at the tertiary level of education in Nigeria. The result revealed that majority of the respondents 30(62.5%) were Christians. The table1 also revealed that majority respondents 46 (95.8%) were single

Tab3: Distribution of Respondents according to their Knowledge on Climate Change

S/N	Items on Knowledge of Climate Change	Response	Frequency	Percentage
1.	Climate change is nothing but a natural fluctuation in temperature of the earth	True	43	89.6
		False	05	10.4
2.	Manifestation of climate change vary from one region to another	True	23	47.9
		False	25	52.1
3.	It is very late for the human community to do anything to stop climate change	True	31	64.6
		False	17	35.4
4.	Agricultural activities like crop and livestock production contribute to climate change	True	27	56.3
		False	21	43.8
5.	`Overall, climate change is bad, it is more harmful than beneficial	True	31	64.6
		False	17	35.4
6.	Human activities do not tangibly affect temperature of the surface of the earth	True	27	56.3
		False	21	43.8
7.	Climate change primarily resulted from ozone hole	True	26	54.3
		False	22	43.8
8.	The sector with the highest contribution to climate change is the transport sector	True	27	56.3
		False	21	43.8

9.	Environmental pollution resulting from the industry is the principal cause of climate change	True	24	50.0
		False	24	50.0
10.	Climate change contributes to increased water and wind erosion of the soil	True	27	56.3
		False	20	41.7
11.	Decline in yield livestock and crop production, thus, deterioration of food security may result from climate change	True	22	45.8
		False	26	54.2
12.	Climate change can result in shortage of water that is suitable for household use, irrigation for crop, livestock use	True	30	62.5
		False	18	37.5
13.	Climate change reduces incidence of infections of animal, plant and human diseases	True	26	54.2
		False	22	45.8
14.	Climate change reduces incidence of contagious animal, plant and human diseases.	True	22	45.8
		False	26	54.2

Source: Field Survey, 2022

Entries in Table 3 show the distribution of respondents according to their Knowledge on Climate Change. The result of Agricultural students' knowledge on climate change as indicated in table 3 revealed that most of the students 43(89.6%) accepted that climate change is nothing but natural fluctuation in temperature of the earth. This is similar to the result of Wang *et al.*, (2019) that the respondents were challenged in understanding data that forms the evidence for global climate change, whether this phenomenon is anthropogenic or natural, or both. Nearly 25(52.1%) of the respondents did not savvy that the manifestations of climate change vary from one climatic region to another, this is contrary to the findings of Filho *et al.*, (2023) that University students agreed that climate change varies from one region to the other, while some of them 17(35.4%) opined that it is very late for human community to do anything to mitigate or control climate

change. More than half 27(56.3%) of the respondents consented that agricultural activities such as crop and livestock production contribute to climate change. Majority of the respondents agreed that human activities tangibly affect temperature of the surface of the earth 27(56.3%) These support findings of Filho *et al.*, (2023) that 70% of the university students attributed the cause of climate change mainly to human activity. whereas 26(54.2%) of the respondents indicated that climate change primarily resulted from the ozone hole. This support the assertion of Ezeudu (2016) that the concentration of the greenhouse gases in the atmosphere caused ozone layer depletion. At the same time, majority 27(56.3%) of the respondents agreed that the transportation sector is the highest contributor to climate change, it was found that the respondents could not decide whether or not environmental pollution resulting from industry is the principal cause of climate change while majority of the respondents 27(56.3) agreed that climate change contribute to increase water and wind erosion of the soil. Whereas 30(62.5%) of the respondents concurred that climate change could result in shortage of water that is suitable for household use, irrigation for crop production and livestock use. The possible explanation for the findings, is that at any rate climate change can result to excessive rainfall that can cause flooding, high rate of run off leading to erosion, it can also be accompanied by high wind velocity that can have destructive effects on human and farming activities. Two, it can result to sudden cessation of rainfall with accompanied drought or shortage of water or suitable water for household use, irrigation (calabash, bucket and watering can irrigation or smallholder farmer irrigation methods) for crop and livestock use. As a final point, the respondents' response 22(45.8%) was that climate change increases incidence of contagious animal, plant and human diseases. This is line with the view of Ekezie (2010) that the rise in temperature as a result of climate change could lead to drought, desertification, declining water table, loss of some plant and animal species, low crop yield and outbreak of climate related diseases like malaria and meningitis. The inference from the mixed responses to items on the knowledge of the respondents on climate change is that the students were not taught any course on climate change. This confirm the issue raised in the problem statement that there is no specific topic on issues of climate change which further implies that students might go through the Agricultural Education programme without adequate or any knowledge of climate change. The result is also in line with findings of Filho *et al.*, (2023) that the student's knowledge of climate change risks varies across gender, age, and academic education. This also corroborate Akrofi *et al.*, (2019) 'students' involvement in climate change-related workshops and campaigns significantly influenced their knowledge levels and students' attitudes and behaviours are dependent on their level of awareness and knowledge about climate change issues. This in turn implies that their cognitive repertoire depends on their level of participation in climate change courses, workshops and campaigns, membership in environmentalist groups, and access to climate information

Table4: Distribution of respondents by their attitudes towards Climate Change

S/N	Items	Response	Frequency	Percentage
1.	Global warming does not give room for concern on climate change	SD	07	14.6
		D	15	31.3
		A	18	37.5
		SA	08	16.6
		SD	18	37.5
2.	The lack of concern for problem of climate change by some individuals, government bothers me	D	11	22.9
		A	05`	10.4
		SA	14	29.2
		SD	13	27.1
		D	07	14.6
3	Scientists will discover solution to climate change, so we need not overly concern ourselves about it	A	10	20.8

4.	Nature has a capacity to heal itself, so I do not worry about climate change	SA	18	37.5
		SD	14	29.2
		D	13	27.1
		A	07	14.6
		SA	14	29.2
5.	Climate change/global warming is a serious problem we are facing now	SD	07	14.6
		D	15	31.2
		A	18	37.5
		SA	08	16.7
		SD	03	6.3
6.	If collective action is taken, climate change effects will be reduced	D	08	16.7
		A	14	29.2
		SA	23	47.9
		SD	05	10.4
		D	13	27.1
7.	Environmental problems associated with global warming can be solved	A	17	35.4
		SA	13	27.1
		SD	13	27.1
		D	13	27.1
		A	10	20.8
8.	Whatever we do, we cannot slow down the warming of the earth due to greenhouse gases	SA	12	25.0
		SD	27	56.3
		D	05	10.4
9.	Man 's activities do not change the climate			

10.	No amount of effort can be put in to change the phenomenon of climate change	A	09	18.8
		SA	07	14.6
		SD	27	56.3
		D	05	10.4
		A	09	18.8
11.	Simple things can be done to slow down global warming or lessen its impact.	SA	07	14.6
		SD	14	29.2
		D	11	22.9
		A	14	29.2
		SA	09	18.8
12.	I plan to take some actions to stop global warming	SD	16	33.3
		D	06	12.5
		A	12	25.0
		SA	14	29.2
		SD	10	20.8
13.	I do not know enough about global warming, so more campaign must be undertaken before action can be taken	D	15	31.3
		A	07	14.6
		SA	16	33.3
		SD	12	25.0
		D	06	12.5
14.	I would like to work with others to slow down global warming	A	20	41.7
		SA	10	20.8

		SD	10	20.8
15.	I am willing to sacrifice my convenience to help reduce greenhouse gas emission	D	13	27.1
		A	12	25.0
		SA	13	27.1
		SD	14	29.2
16.	My activities have no significant contribution to global warming, so I do not intend to take actions to change them	D	15	31.3
		A	12	25.0
		SA	07	14.6

SD: Strongly Disagree, D: Disagree, A: Agree, SA; Strongly Agree

Field Survey, 2022.

Entries Table 4 show the distribution of respondents by their attitudes towards Climate Change. The result shown that majority of the respondents 18(37.5%) indicated that global warming did give room for concern on climate change, This implies the issue of global warming should give everybody cause for concern because there is gradually reduction in ozone layer which bring about various effects of climate change being experienced in world today, while 18(37.5%) on the contrary indicated that the lack of response by individuals and government towards the problem of climate change did not bother them. Majority 18 (37.5%) opined that scientist will discover solutions to the problem of climate change, most of them 18(37.5%) concurred that climate change/global warming is a serious problem we are facing now, 23(47.9%) of them strongly agree that if there is collective action, we will be able to reduce the effects of climate change. About 17(35.4%) of the respondents agreed that they will be able to solve environmental problems associated with global warming. Most of the respondents 13(27.1%) strongly disagree that whatever they do, they cannot slow down the warming of the earth due to greenhouse gases. It means if appropriate actions are taken by stakeholders, greenhouse gas emission may be controlled. Majority of the respondents 27(56.3%) strongly disagree that man's activities do not change the climate as well as no amount of effort can be put in to change it. It means human activities contribute to the problem of climate change and if appropriate climate smart programmes are put in place the phenomenon could be reversed particularly in relation to Agriculture or farming activities. Nearly 16(33.3%) plan to take some actions to stop global warming while 20(41.7%) would like to work with others to help slow down global warming. These imply that climate change mitigation and adaptation strategies should be a collective action to be undertaken by all parties involved in climate change actions. Some of the respondents 16(33.3%) concurred, not knowing enough about global warming, so more campaign must be undertaken before taken action. This emphasizes the need for the inclusion of climate change courses in the curriculum at the NCE programme so the effects can trickle down to the Basic level of education (i.e. Primary and Junior secondary schools). Also, sensitization on causes, effects and mitigation and adaptation strategies on climate change at the farm level. All of the above responses of the respondents are in line with the views of Shaman and Knowlton, (2017) that students' involvement and participation in climate change adaptation and mitigation activities are crucial to address the direct and indirect impacts of climate change. However, since students have not reached the desired level of consciousness, educational curriculums on climate change need to be designed to positively affect students' perceptions about the environment, and their courses should improve university students 'or tertiary institution students' scientific skills and knowledge that shape their attitudes and beliefs about climate change. This finding supports Oruonye (2011) who found out that tertiary institution students possessed low attitude towards climate change. Ishaya and Obaje (2008) also found out

that secondary school students in Jema, Kaduna State, possessed low attitude towards climate. Majority of the respondents disagree 15(31. 3%) with the notion that their activities have no significant contribution to global warming so I do not intend to take actions to change them. It means that their activities contribute significantly to the problem of global warming and they are willing to take action to change the situation. The inference that can be drawn from the responses of the respondents above is that if the students were taught, given adequate orientation about the phenomenon of climate change at the NCE level or whatever level of education they will be able to tackle the scenario of climate change in the future and they are also willing to take positive actions to solve the problem of climate change. This is line with the view of Bozoglu *et al.*, (2022) that because these students will either be part of experts or general public in near future. This is corroborated by Molthan-Hill *et al.* (2019), that the role of the University (tertiary institutions) is important in climate change education in order for the future leaders to make informed decisions and for the public needs to be embedded in the climate change mitigation and adaptation tools and practices.

Table5: Correlation between socio - economic characteristics of respondents and their knowledge of climate change

Socio- economic characteristics	Correlation	Students' knowledge
Age of respondents	Pearson Correlation	.241
	Sig. (2-tailed)	.099
	N	48
Respondent Gender	Pearson Correlation	-.054
	Sig. (2-tailed)	.715
	N	48
Respondents Tribe	Pearson Correlation	.110
	Sig. (2-tailed)	.458
	N	48
Level of study	Pearson Correlation	.268
	Sig. (2-tailed)	.066
	N	48
Respondents Religion	Pearson Correlation	.221
	Sig. (2-tailed)	.131
	N	48
Marital Status	Pearson Correlation	.071
	Sig. (2-tailed)	.631
	N	48

	Pearson Correlation	1
Students' knowledge	Sig. (2-tailed)	.240
	N	48

$P > 0.05$ = Not significant but if $P < 0.05$ = shows that there is significant relationship

Source: Field Survey, 2022

The results in Table 5 show the correlation coefficients of the relationship between selected socio-economic characteristics and agricultural students' knowledge of climate change. There was no relationship between the selected socio-economic characteristics and students' knowledge of climate change such as age of respondents (.241, $P > 0.05$). Respondents gender (- 0.54, $P > 0.05$). Tribe (.110, $P > 0.05$). Level of study (.268, $P > 0.05$). Respondents Religion (.221, $P > 0.05$) and Marital Status (.071, $P > 0.05$). This implies that no topic on climate change was included in the Agricultural education curriculum as the students did not have any knowledge of climate change. This confirms the issue raised in the problem statement that there is no specific topic on issues of climate change which further implies that students might go through the Agricultural Education programme without adequate or any knowledge of climate change. The result is also in line with findings of Filho *et al.*, (2023) that the student's knowledge of climate change risks varies across gender, age, and academic education. This also corroborates Akrofi *et al.*, (2019) 'students' involvement in climate change-related workshops and campaigns significantly influenced their knowledge levels and students' attitudes and behaviours are dependent on their level of awareness and knowledge about climate change issues. This in turn implies that their cognitive repertoire depends on their level of participation in climate change courses, workshops and campaigns, membership in environmentalist groups, and access to climate information.

Table 6: Correlation between socio - economic characteristics of respondents and their Attitude towards climate change

Socio - economic characteristics	Correlation	Students Attitude
Age of respondent	Pearson Correlation	.178
	Sig. (2-tailed)	.226
	N	48
Respondents Gender	Pearson Correlation	-.173
	Sig. (2-tailed)	.239
	N	48
Respondents Tribe	Pearson Correlation	-.041
	Sig. (2-tailed)	.785
	N	48
Level of Study	Pearson Correlation	.330*

	Sig. (2-tailed)	.022
	N	48
	Pearson Correlation	_.082
Respondents Religion	Sig. (2-tailed)	.581
	N	48
	Pearson Correlation	_.026
Marital Status	Sig. (2-tailed)	.859
	N	48
	Pearson Correlation	1
Students Attitude	Sig. (2-tailed)	
	N	1104

Source: Field Survey, 2022 $P > 0.05$ = Not significant but if $P < 0.05$ = Significant

Table 6 also revealed that most of the respondents' socio-economic characteristics did not have any significant relationship with their Attitude towards climate change except their levels of study which show a significant relationship ($r = .330^*$, $P < 0.05$) with their Attitude. The inference that can be drawn from the responses of the respondents above is that if the students were taught, given adequate orientation about the phenomenon of climate change at the NCE level or whatever level of education they will be able to tackle or take proactive actions against the scenario of climate change in the future. The findings of Filho *et al.*, (2023) pointed out that universities (tertiary institutions) could improve student awareness by organising climate change-related themes in discussion sessions and incorporating the outcome into the curriculum and co-curricular activities.

Conclusion

Based on the results of this study, it can be concluded that respondents' socio-economic characteristics had no significant relationship with their attitudes as well as their knowledge of climate change except their levels of study which had significant relationship with their attitudes toward climate change.

Recommendations

Based on the results of this study, the following recommendations are hereby made:

1. Curriculum at the National Certificate of Education should be reviewed by making climate change part of the topic in Agro-climatology course which must be offered by all the students irrespective of their course of study in order to enlighten them on the effect of climate change.
2. Climate change awareness club should be established in higher institutions so as to improve students' knowledge and attitudes toward climate change, such clubs can be challenged to identify and find solution to the insurmountable problems of climate change.

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